MIL-T-47014(AR)
19 April 1974
SUPERSEDING
MIS-14365
21 July 1966

MILITARY SPECIFICATION

TAPE, ALUMINUM FOIL, PRESSURE-SENSITIVE,

HEAT REFLECTIVE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of heat reflective aluminum foil pressure-sensitive tape intended for use at elevated temperatures.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated Triple-Wall

Military

MIL-P-116	Preservation, Methods of
MIL-A-148	Aluminum Foil

FSC 8030

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STANDARDS

Federal

FED-STD-147	Tape, Pressure Sensitive and Gummed, Methods of Inspection and Testing

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (6.2), a preproduction sample of tape shall meet the requirements of this specification.

3.2 Material. The tape shall consist of 0.002 plus or minus 0.001 inch thick aluminum foil coated on one side with a high temperature pressure-sensitive adhesive. Unless otherwise specified (see 6.2), the width and length shall be as specified in the contract or order.

3.3 Mechanical properties.

3.3.1 Tensile strength. The tensile strength of the tape shall be not less than 14 pounds per inch of width.

3.3.2 Adhesion.

3.3.2.1 After room temperature exposure. Under ambient temperature exposure, the tape shall have an adhesion level of not less than 15 ounces per inch of width.

3.3.2.2 After high temperature conditioning. Tape, aged at plus 260 plus or minus 5.5 degrees Celsius (C) (plus 500 plus or minus 10 degrees Fahrenheit (F)) for 60 plus or minus 5 hours shall have an adhesion level of not less than 10 ounces per inch of width, measured at room temperature.

3.4 Physical properties.

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3.4.1 Thickness. The total thickness of the tape shall be not greater than 0.0045 inch.

3.4.2 Surface finish of foil. The surface finish of the aluminum foil shall be either dry annealed or slick annealed.

3.4.3 Contamination. The tape shall be free of dirt, foreign material, or other contaminants.

3.5 Shelf life. The shelf life of the tape in the packaged condition shall be not less than 1 year from date of manufacture when stored at a temperature less than plus 27 degrees C (plus 80.6 degrees F).

3.6 Workmanship. The workmanship shall be such as to insure a high quality product which is uniform and in conformance with this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of tape. Preproduction samples which do not meet all of the requirements of this specification shall be subject to rejection and returned to the supplier.

4.3 Classification of tests. The examination and testing of the tape shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

4.3.1 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein.

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4.3.2 Quality conformance tests. Quality conformance tests for acceptance of the tape shall consist of the following examinations and tests:

a. Visual examination for surface finish and contamination.

b. Total thickness of the tape.

c. Thickness of the aluminum foil.

d. Adhesion level without high temperature conditioning.

4.3.2.1 Lot size. Lot size shall consist of all the tape manufactured at one time, from one batch, forming part of one contract or order, and submitted at the same time and place.

4.3.2.2 Sampling. Unless otherwise specified (see 6.2), tape representative of each lot shall be selected at random for quality conformance testing. Each roll of tape shall be considered as a unit of product. Failure of sample to meet the quality conformance requirements shall be cause for lot rejection.

4.4 Test methods.

4.4.1 Mechanical properties.

4.4.1.1 Tensile strength. The tensile strength shall be determined in accordance with FED-STD-147, Method 30. The tensile strength shall be as specified in 3.3.1.

4.4.1.2 Adhesion. Adhesion shall be determined in accordance with FED-STD-147, Method 10, using test specimens 10.000 plus or minus 0.125 inch long and 1.000 plus or minus 0.032 inch wide. The adhesion level shall be as specified in 3.3.2.1 and 3.3.2.2.

4.4.2 Physical properties.

4.4.2.1 Total thickness. The thickness of the tape, including foil and adhesive, shall be determined by a micrometer or other suitable method. The total thickness shall be as specified in 3.4.1.

4.4.2.2 Foil thickness. The foil thickness shall be determined by separating the adhesive from the foil and measuring the foil with a micrometer or other suitable method. The foil thickness shall be as specified in 3.2.

4.4.2.3 Surface finish of foil. The surface finish of the aluminum foil shall be determined in accordance with MIL-A-148. The surface finish shall be as specified in 3.4.2.

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4.4.2.4 Contamination. The tape shall be visually examined for dirt, foreign material, or other contaminants.

4.4.3 Shelf life. The tape shall be tested for conformance with all requirements specified herein after completion of the storage period specified in 3.5. The supplier's certificate of compliance with 3.5 may be accepted during the storage period.

4.5 Preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking shall be examined to determine compliance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified (see 6.2), preservation, packaging, packing, and marking shall be as specified herein.

5.2 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

5.2.1 Level A. Packaging of the tape shall be in accordance with MIL-P-116, Method III. Rolls of tape shall be inserted into weather resistant containers conforming to PPP-B-636.

5.2.2 Level C. Tape shall be preserved and packaged to afford adequate protection against deterioration and damage during shipment from supply source to first receiving activity.

5.3 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.3.1 Level A. A specified quantity of tape packaged as specified shall be packed in a snug-fitting shipping container conforming to PPP-B-640, Class 2, or a weather resistant container conforming to PPP-B-636.

5.3.2 Level B. Same as Level A.

5.3.3 Level C. Tape packaged as specified, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with the rules and regulations of the carrier applicable to the mode of transportation.

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Materials (ASTM) Publications:

American Society for Testing and

5.4 Marking. In addition to any special markings required by the contract or purchase order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tape is intended for use as a heat reflective shield for extended operation up to plus 260 degrees C (plus 500 degrees F) and short time exposure up to plus 538 degrees C (plus 1000 degrees F).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required, and, if so, pertinent details (see 3.1).

c. Sampling plan if other than specified (see 4.3.2.2).

d. Requirements for preservation, packaging, packing, if different (see 5.1 and 5.2).

e. Applicable level(s) of preservation, packaging, and packing (see 5.2 and 5.3).

f. Width and length required (see 3.2).

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification, MIS-14365, dated 21 July 1966.

Custodian:
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Preparing Activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-47014	TAPE, ALUMINUM FOIL, PRESSURE-SENSITIVE, HEAT REFLECTIVE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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